Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/24/2008
PI Name:	Best, Fredrick Ph.D.		
Project Title:	Vortex Dehumidification System		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	College Station	**State:**	TX
Zip Code:	77843-3133	**Congressional District:**	17
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	01/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: changed end date to 09/30/2007 per info from CoI on extensions granted (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kurwitz, Richard (Texas A&M University)		
Grant/Contract No.:	NNJ04HD74G		
Performance Goal No.:			
Performance Goal Text:			
	Dehumidification in air handling systems aboard the ISS and future space vehicles is problematic and not efficient. The proposed project would be to demonstrate that utilizing the Texas A&M Vortex Separator (part of the Immobilized Microbe Microgravity Wastewater Processing System (IMMWPS) Flight Unit) as part of a dehumidification system for future spacecraft or as an ISS replacement unit is feasible. The vortex separator would be chilled to low temperature and ambient air would be injected through the device. Air leaving the device would be saturated at the chilled temperature and then heated to ambient conditions reducing the relative humidity. A further feature of the system is that it could potentially be used in other gas/liquid separation such as microgravity showers or other high gas volume flows to reduce the entrained liquid. It is expected that the vortex system will reduce power requirements and relative mass/volume while increasing reliability.		

Task Description:	The ITP plans to accomplish the following main objectives: • Demonstrate, through microgravity experimentation, the capability of the separator to provide a passive solution to cabin air dehumidification that is simpler, more adaptive, and less prone to failure than current mechanically driven technology. • Thoroughly test and optimize a modified separator design and flow loop for this application. Currently, dehumidification systems utilize mechanically driven separators. Due to the nature of these separators, moving parts, maintenance, and therefore additional modes of failure are introduced into these systems. While these statements are true for any mechanical system, the current systems have proven them true through failure. In January 2003, the primary rotary separator on the Spacehab module failed due to design limitations that will be discussed shortly. The secondary separator was brought on-line, but failed due to mechanical problems, resulting in complete failure of the dehumidification system. This failure delayed experiments and produced uncomfortable living conditions for the crew of Spacehab. Mechanical failure can be avoided by replacing the rotary separator with a passive separator. Using the intrinsic momentum of the fluids, whether two-phase, all gas, or all liquid, the ITP separator drives the buoyancy process with no moving parts, thereby significantly reducing the opportunity for failure. This is an important and attractive advantage of passive separators. Secondly, the design of current rotary separators is sensitive to large variations in flow conditions. The primary rotary separator of the Spacehab system failed because the design could not accommodate the flow conditions, resulting in flooding of the cabin air conditioning system. One of the aspects of the ITP design is the unique ability to accommodate significant flow variations. In effect, this separator acts as an accumulator for the system, handling system volume variations and eliminating the need for a separate component to do so. While the possibility of flooding or dry-out does exist, the hydrodynamic limits resulting in such failure have been studied extensively by the ITP and are well known. For this reason, the separator can be designed to handle any flow variations required by the system. This reduces the sensitivity of the system to flow changes and further reduces modes of failure. Moreover, the ITP has recently completed development of a dynamic volume sensing technique that allows control software to monitor and adjust microgravity flow conditions without operator interaction. This technique increases both the reliability and efficiency of the accumulation function associated with the ITP separator. Aside from significantly reducing the probability of failure, the simplicity of a passive separator also reduces required system space, mass, and power. The ITP has achieved successful separation with driving flows of 4 cfm using a 4.5 inch diameter, 4.5 inch tall separator. It is important to note that the maximum air flow rate for this size has not been tested, but the tested flow is comparable to the 6.3 cfm currently processed by the Common Cabin Air Assembly (CCAA) of the International Space Station (ISS). Furthermore, the separator size can be optimized to accommodate air flow ranges required by cabin air conditioning systems. In conclusion, the passive vortex separator developed by the ITP has the potential to significantly reduce failure modes, required space, system mass, and power requirements for cabin air conditioning systems. The simplified designed relates to less crew time spent on maintenance, while failure reduction minimizes crew time spent in undesirable living conditions. All of these factors indicate the passive approach to cabin air dehumidification provide positive answers to the questions posed by the AHST program, as well as a method to better optimize current cabin air conditioning systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The passive vortex separator developed by the ITP has the potential to significantly reduce failure modes, required space, system mass, and power requirements for cabin air conditioning systems. The simplified designed relates to less crew time spent on maintenance, while failure reduction minimizes crew time spent in undesirable living conditions. All of these factors indicate the passive approach to cabin air dehumidification provide positive answers to the questions posed by the AHST program, as well as a method to better optimize current cabin air conditioning systems.
	The Vortex Dehumidification System (VDS) developed at Texas A&M University (TAMU) was successfully tested aboard reduced gravity aircraft. These tests showed the Vortex Separator (VS) to be more than capable of dehumidifying and cooling prototypic ISS inlet air streams by maintaining the liquid vortex at a temperature below the dew point of the incoming humid air. Furthermore, the research demonstrated the ability to control the outlet air conditions through the temperature of the liquid within the separation volume. Regardless of inlet air conditions, the air stream reached equilibrium with the liquid before exiting the VS. The VS demonstrated humidity removal rates in excess of 16 grams of water per cubic meter of air (16g/m3). The TAMU vortex separator dehumidification system can replace the current CCAA system on the ISS or be considered for future spacecraft due to two primary advantages: Mass and Volume are reduced and performance is improved. Based on the preliminary trade study shown in Section 6, the TAMU vortex separator dehumidification system requires only a third of the mass and half the volume of the existing system without a loss in performance. The power required is significantly decreased due to the removal of a powered rotary separator. Maintenance is simplified and the need for consumables is reduced. Compared with other separators, there are no high precision or high speed moving parts which can cause vibration or require frequent maintenance and there are no capillary pores to be plugged. Since the separator catches dust in a liquid layer, no separate filters are required. The condensation rate is a result of the difference in the inlet air stream properties and the equilibrium condition maintained by the liquid temperature. For air of consistent inlet temperature and humidity, condensation rates can be increased by lowering liquid temperature or increasing the inlet flow rate. As inlet temperature, humidity, and flow rate levels fluctuate, the condensation rate will change to maintain a stable outlet temperature and humidity level. The outlet flow rate will be equivalent to the inlet flow rate less the condensation rate.

Task Progress:	Furthermore, the separation rate always matches the condensation rate as these two processes are essentially one. In other words, since condensation occurs at the bubble interface, any condensate that forms is simultaneously separated to the liquid vortex. This is the major advantage of direct contact heat exchange over plate type heat exchangers. For the latter type, condensate accumulates on the heat transfer surface. With current technologies, this condensate is either removed from the surface through capillary forces to a low pressure sump, such as with a porous heat exchanger, or sheared by the air flow through a slurper type separator and directed to a rotary separator for final separation. If the condensation rate overcomes the separation or removal rate, condensate builds up on the heat exchanger surface and increases the thermal resistance between the air flow and low temperature plate, reducing thermal energy transfer. Hydrophobic and hydrophilic coatings have been developed to combat this issue but cannot completely overcome this effect. In addition, these treated surfaces are subject to microbial growth and fouling of the special coatings, resulting in degraded performance over time. By using direct contact heat exchange, the VDS does not encounter these problems. The ability of the VDS to maintain a stable outlet condition despite varying inlet conditions is due to the large heat transfer area of the device. This area is large due to the direct contact heat exchange method and the unique operating characteristics of the VS. Inlet gas is quickly broken up into bubbles on the order of a millimeter in diameter by the high velocities present in the liquid vortex near the injection point. The surface area of these bubbles provides the direct contact heat exchange area. As the TAMU VS is capable of handling large inlet volumetric flow rates, this heat exchange area is considerably large compared with the overall volume of the system. For example, a 4.5 inch diameter, 4.5 inch tall separator can have heat transfer areas of several square meters. Furthermore, testing showed that, for higher than normal cabin air conditions approaching 120 degrees Fahrenheit, at most 30% of the available heat transfer area was required to reach thermal equilibrium between the inlet air stream and liquid vortex. For typical cabin air conditions, less than 10% of the available area is required. This indicates that the VDS is far from maximum capacity and could handle considerably larger flow rates than the maximum tested, 120 SLPM. Another unique aspect of the VDS is that the heat transfer area is linearly proportional to the inlet flow rate. As a result, as demand increases and cabin air flow rates increase, the available heat transfer area increases passively. This feature allows the VDS to handle large and sudden changes in inlet volumetric flow rate with flow rate being the only change in outlet air conditions. This is a considerable performance advantage over the current rotary systems, which have shown to fail during sudden or large inlet flow rate transients. The research conducted by TAMU and presented in this document successfully demonstrated that the VDS is more than capable of providing cabin air dehumidification in reduced gravity environments. Unlike current approaches, this system does not require a separate condensing heat exchanger and separation device but combines phase separation and dehumidification into a single unit. Furthermore, the VDS is not sensitive to inlet flow rate, temperature, or humidity variations and, during these transients, passively adjusts to produce a constant air quality at the outlet. As shown through experiment, a VDS operating with a liquid temperature of 40 degrees Fahrenheit will produce a stable outlet air stream with a vapor content 8.4 grams of water per cubic meter of air. Upon remixing with cabin air at 72 degrees Fahrenheit, this equates to a relative humidity of 33%.
Bibliography Type:	Description: (Last Updated: 03/24/2008)
Papers from Meeting Proceedings	Williams R, Carron I, Bray D, Kurwitz C, Best F. "Zero-gravity test results for ultrasonic sensing of air-liquid interface in a vortex separator." Space Technology and Applications International Forum-STAIF 2003: Conf. on Thermophysics in Microgravity; Commercial/Civil Next Generation Space Transportation; Human Space Exploration; Symps. on Space Nuclear Power and Propulsion (20th); Space Colonization (1st). January 2003. AIP Conference Proceedings, vol. 654, 2003. p. 239-246. http://dx.doi.org/10.1063/1.1541299 , Jan-2003
Papers from Meeting Proceedings	Kurwitz C, Ellis M, Best F. "Reduced Gravity Direct Contact Heat Exchanger for Cabin Air Dehumidification." 11th Biennial ASCE Aerospace Division International Conference - Engineering, Science, Construction and Operations in Challenging Environments - Earth and Space Conference 2008, Long Beach, California, March 3-5, 2008. 11th Biennial ASCE Aerospace Division International Conference - Engineering, Science, Construction and Operations in Challenging Environments - Earth and Space Conference 2008, Long Beach, California, March 3-5, 2008. , Mar-2008